

1. Title

Predicting Pokémon VGC Battle Outcomes from Team Composition and Starting Leads: a Machine Learning Approach

2. Abstract

This research project dives into displaying the potential of matchup knowledge using machine learning. Pokémon battles by design require the player to know about all of the battle-defining information of each individual Pokémon (on both their own and the opponent's team) in order to secure a win. Consequently, a highly skilled player, simply by just looking at what Pokémon opposing teams have, and what starting leads each player chooses, can understand who has a higher chance of winning. However, players who are less so certainly do not have that understanding; this is the research problem the project tackles with the help of ML. Our question is: "given two opposing team compositions and their starting leads, can we determine which team is more likely to win?" The study will read team and starting-lead data of online battles and use them to train several supervised models. Model performance will be compared using held-out test data to determine how effectively pre-battle information can predict battle outcomes. This project is expected to contribute a reproducible machine learning pipeline for Pokémon VGC battle prediction and a comparison of various ML approaches. The project will examine the predictive value of team composition and starting-lead selections, analyze model errors/uncertainty, and point out limitations that come with using online competitive battle data.

3. Research Background

Competitive video games are a contest of strategy, skill, practice, and game knowledge, as well as natural talent and luck. The eSport this project is concerned with, Pokémon VGC (Video Game Championships), is a prime example of strategy-heavy competition in eSports. It has seen substantial growth in its competitive scene in 2026 following the release of Pokémon Champions in April, which amassed 20 million downloads worldwide by August (Kim, 2026). Additionally, the Pokémon World Championships, VGC's most prestigious tournament series, reached a new peak viewership in 2026 with over 200,000 concurrent viewers during the grand final (Esports Charts, n.d.). With this newfound popularity, competitive Pokémon reach new limits with many people around the world helping define the metagame, refine compositions and strategies, or just increase their own knowledge of matchups. Knowledge of matchups is arguably the limiting factor of skill expression for most Pokémon players, as the amount of variety in team composition in the game causes matchups to be extremely dynamic in nature; it is rarely a discrete "this" or "that" when it comes to which team composition will win.

The goal is to knock out your opponent Pokémon before they knock out yours. Each player has a team of 6 Pokémon and can only bring 4 to a match. A match only has 2 playable pokemon for each

player at a time while the other 2 are in the players backline and can be switched in. Players give commands simultaneously but the order of Pokémon actions is reliant on their speed stat.

The lead in Pokémon VGC is the 2 Pokémon a player decides to lead with and is arguably the most important turn of match in Pokémon VGC. A favourable lead gives a player an advantage that can snowball for the rest of the game, ultimately crowning them the victor of the match. We would like to predict the winner of a Pokémon VGC match based on the pokemon each player leads with.

Current research struggles with the high dimensionality of Pokémon VGC. Just considering lead combinations of both players gets to a number of 255 combinations. On top of this each pokemon has 4 moves, an item slot, and an ability which is a passive effect. According to a research paper on VGC-Bench (a benchmark for AI playing VGC), current algorithms start to degrade when scaling to multi team generalization due to the combinatorial complex environment of VGC. By focusing on leads we can get insights on how leads can determine the match while reducing dimensionality.

5. Methodology

Pokémon Champions, the current standard video game for competitive Pokémon VGC battling, uses a season-based regulation ruleset which defines what Pokémon, items (which are used by Pokémon), and moves are legal in battle. So far, 3 different regulations have been released in the game's lifetime: *M-A*, *M-B*, and *M-C*. The first two lasted sequentially from the release of the game until September 9, 2026, which was when *M-C* took over (Victory Road, 2026). The dataset we plan to utilize is a JSON table containing battle data of online matches held during *M-A* and *M-B*, so our model will only be looking at Pokémon matchups that conform to those regulations (not all existing Pokémon/items/moves will be present). The matches themselves were scraped from an online Pokémon battle simulator, called *Pokémon Showdown*; 88,905 total battles were scraped from the simulator website's replay database (Angliss, 2025).

Several ML approaches will be implemented, tested and compared. A naive baseline will set the minimum level of performance expected from any predictive model. Logistic regression will be used as an interpretable baseline, while decision trees and random forests will be investigated as models for nonlinear relationships between team compositions and matchup characteristics. Feature sets will also be compared to determine whether starting-lead information provides additional predictive value beyond team composition alone.

The ML models will be evaluated with a consistent training, validation, and test procedure. Hyperparameters will be selected based on validation data (the regulation *M-B* datatable) while the final test set will be unseen to the model until final evaluation. For presentation/demonstrative purposes, the model will also be tested on a popular matchup that took place at the recent 2026 Pokémon World Championships—Wolfe Glick vs. Hirofumi Kimura.

Performance will be measured primarily using accuracy, as it will determine the ratio of battles predicted correctly to incorrectly, and is easily digestible to view. Other metrics such as precision, recall, and F1-score will help show reliability of the models and their classifications. ROC-AUC may be utilized to help us evaluate how well a model ranks one team's wins above an opposing team across different probability thresholds; this will give us a different perspective compared to F1-scores.

The main comparison we want to make is whether more complex models give us improved predictive performance over the logistic regression and naive baselines, and whether incorporating starting-lead information (what two Pokémon opposing players will choose to start the battle with) improves predictions. Error analysis will look at cases in which predictions differ from the actual outcome of the battle, particularly in uncertain matchups or “mirror” matchups (in which opponents have similar team compositions, starting leads, or both).

6. Significance

Any action in Pokemon VGC is highly volatile meaning any advantage can snowball in one's favour. A model predicting who wins based on team leads provides further insight on how leads determine the outcome of a match. Other machine learning researchers could further expand upon this idea to find models that find the best move or evaluate the current board state, similar to chess. This would be a boon for competitive players by allowing them to understand VGC gamestates at a deeper level. Newer players would stand to benefit from this as the model could give information on the skill of choosing leads, which is more prevalent than ever because of the uptick of new players due to Pokemon Champions releasing in early 2026. This research could also be used by casters and organizers could use this information for overlays to enhance the viewing experience and give new players a more in-depth understanding of strong moves.

7. SMART Timeline

By Friday, October 2nd, our group will have confirmed our data sets, completed a data card, and executed an initial Exploratory Data Analysis. This allows us to establish a clear training, validation, and testing strategy for our model.

By Monday, October 19th, our group will have delivered a working data pipeline, deployed a benchmark model, created documentation to allow for a fully reproducible environment allowing for known prediction discrepancies.

By Friday, October 30th, our group will have given the current methods we are using towards our model, produced a theoretical justification and compile preliminary model results, and present any necessary timeline or blocker revisions/resolutions by the end of our midpoint check in.

By Friday, November 20th, our group will provide fair comparisons of all our required models, execute

system tuning, and perform a sensitive robust ablation study alongside a detailed error analysis with drafts of our limitations and conclusions.

By Friday, December 4th, our group will have prepared an 8 - 10 minutes presentation of our model alongside a detailed 6-8 page final report, a fully reproducible code repository and a fully honest AI use statement, and positive response to feedback on our model.

By Monday, December 7th, our group will have prepared a 1 - 2 page honest individual reflection with evidence of contribution and key theoretical/experimental decisions, documentation of one failure and or lesson learned during the project, a fully honest AI use statement and a 5 minute code walkthrough.

8. References

- Angliss, C. (2025, May 13). *cameronangliss/vgc-battle-logs · Datasets at Hugging Face*. Hugging Face, Inc. Retrieved September 90, 2026, from <https://huggingface.co/datasets/cameronangliss/vgc-battle-logs>
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- Esports Charts. (n.d.). *2026 Pokémon World Championships - VGC - Pokémon - Viewership, Overview, Prize Pool | Esports Charts*. Esports Charts. Retrieved September 20, 2026, from <https://escharts.com/tournaments/pokemon/2026-pokemon-world-championships-vgc>
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- Victory Road. (2026, April 8). *Pokémon Champions — Regulations*. Victory Road. Retrieved September 20, 2026, from <https://victoryroad.pro/champions-regulations/>